DRUGS · ANTIBIOTICS: PENICILLINS

# Phenoxymethylpenicillin and benzylpenicillin

Phenoxymethylpenicillin (penicillin V) and benzylpenicillin (penicillin G) are the original narrow-spectrum penicillins. Group A streptococcus has never developed resistance to them, which makes them the first-line drugs for streptococcal tonsillitis and pharyngitis, and, in NICE guidance, for acute sinusitis in adults. Penicillin V is the oral form; benzylpenicillin is given intravenously to patients who cannot swallow, with severe tonsillitis or quinsy.

## 1 CHAPTER ONE At a glance

The drugs on one page.

### 1.1 Summary

PHENOXYMETHYLPENICILLIN AND BENZYLPENICILLIN

| | SUMMARY |
|---|---|
| **Class** | Natural penicillins (beta-lactam) |
| **How they work** | Inhibit cell wall cross-linking; bactericidal, time-dependent |
| **Spectrum** | Streptococci (group A, C, G and pneumococci), many oral anaerobes, Neisseria, Treponema, Actinomyces; not most staphylococci, not H. influenzae or Moraxella that produce beta-lactamase, not Gram-negative bacilli |
| **ENT uses** | Streptococcal tonsillitis and pharyngitis (penicillin V); acute sinusitis in adults (NICE first choice); severe tonsillitis, quinsy (with metronidazole) and actinomycosis (benzylpenicillin); scarlet fever; prophylaxis after splenectomy |
| **Adult dose** | Penicillin V 500 mg four times daily (or 1 g twice daily for sore throat), up to 1 g four times daily; benzylpenicillin 1.2 g four times daily intravenously, up to 2.4 g every 4 hours in severe infection |
| **Child dose** | Penicillin V 1 to 11 months 62.5 mg, 1 to 5 years 125 mg, 6 to 11 years 250 mg, 12 to 17 years 500 mg, four times daily; benzylpenicillin 25 mg/kg every 6 hours (50 mg/kg in severe infection) |

| | SUMMARY |
|---|---|
| **Duration** | Sore throat 5 to 10 days (10 days is still advised in many guidelines to eradicate the streptococcus); sinusitis 5 days |
| **Avoid** | Penicillin allergy |
| **Pregnancy** | Safe |

# 2 CHAPTER TWO How they work

Mechanism and handling.

## 2.1 Mechanism

Like all penicillins they bind penicillin-binding proteins and stop peptidoglycan cross-linking. Their narrow spectrum is an advantage: less disruption of the gut and oral flora, less C. difficile and less resistance selection.

## 2.2 Absorption and handling

Penicillin V is acid-stable and absorbed by mouth but best taken **on an empty stomach**, an hour before food. Benzylpenicillin is destroyed by gastric acid and must be given parenterally. Both have short half-lives (about 30 to 60 minutes), hence dosing four times a day; probenecid prolongs their levels.

# 3 CHAPTER THREE Using them in ENT

Indications and doses.

## 3.1 Sore throat

Use a clinical score to decide: **FeverPAIN** (fever, purulence, attend rapidly within 3 days, inflamed tonsils, no cough or coryza) or **Centor** (exudate, tender anterior nodes, fever, absent cough). With FeverPAIN 0 to 1 or Centor 0 to 2, no antibiotic;

FeverPAIN 2 to 3, consider a back-up prescription; FeverPAIN 4 to 5 or Centor 3 to 4, consider an immediate antibiotic. Penicillin V is first-line; clarithromycin or erythromycin in allergy.

### 3.2 Severe tonsillitis and quinsy

A patient who cannot swallow fluids or has a peritonsillar abscess is admitted: **intravenous benzylpenicillin** (often with **metronidazole** for anaerobes), fluids, analgesia, and drainage of any abscess; many units give a single dose of dexamethasone. Step down to oral penicillin V (with metronidazole) once swallowing.

### 3.3 Other uses

- **Actinomycosis** of the jaw and neck: high-dose benzylpenicillin for weeks, then oral penicillin for months.
- **Acute sinusitis in adults** where an antibiotic is indicated.

## 4 CHAPTER FOUR Safety

What goes wrong.

### 4.1 Adverse effects

- Hypersensitivity: from rash to anaphylaxis; the commonest cause of drug anaphylaxis.
- Diarrhoea and nausea, less than with broader agents.
- Very high doses of benzylpenicillin in renal failure can cause **seizures** (encephalopathy).
- The **Jarisch-Herxheimer reaction** in syphilis.

## 4.2 Contraindications and cautions

- Penicillin allergy. About 10% of patients carry a label, but most are not truly allergic; formal de-labelling widens safe options.
- Renal impairment with high-dose intravenous benzylpenicillin.

## 4.3 Interactions

Methotrexate (reduced excretion); probenecid (raised levels).

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **FeverPAIN** and **Centor** scores and how they guide antibiotics.
- Why penicillin V, not amoxicillin, is first-line for a sore throat.
- Treatment of **quinsy**: drainage plus intravenous benzylpenicillin and metronidazole.
- Group A streptococcus remains universally penicillin-sensitive.

## 5.2 Common pitfalls

COMMON PITFALLS

- Prescribing penicillin V to be taken with meals.
- Short courses that fail to eradicate streptococcus in a patient with recurrent tonsillitis.
- Relying on antibiotics alone for a peritonsillar abscess.

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